

PART I - FOUNDATION PLC PROJECTS

Project 10 - Industrial Packaging and Sorting Station

PLC Platform: Allen-Bradley MicroLogix 1100

Programming Software: RSLogix Micro Starter Lite 8.30

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1. Project Overview

The Industrial Packaging and Sorting Station was developed as the final Version 1 PLC project in this portfolio. It combines maintained system operation, conveyor control, package counting, batch-complete handling, reject detection, a timed reject-diverter cycle, reject counting, operator indication, and reset logic.

Start Push Button I:1/0 latches System_Run B3:0/0 while Batch_Complete B3:0/2 is false. While the system is running and the batch is incomplete, Conveyor Motor O:2/0 and System Running Lamp O:2/3 are energized. Package Photoeye I:1/2 increments Package Counter C5:0, which has a preset of 10.

When C5:0 reaches its preset, Batch_Complete is latched. This stops the conveyor, turns off the running indication, energizes Batch Complete Lamp O:2/4, and energizes Buzzer O:2/2 until reset.

The reject path operates independently from the batch-complete indication while the system is running. Reject Sensor I:1/3 latches Reject_Cycle_Active B3:0/1, energizing Reject Diverter O:2/1 for a timed two-second cycle. When the reject timer completes, the reject cycle is cleared and Reject Counter C5:1 increments.

2. Engineering Objective

The objective of this project was to integrate several previously learned PLC concepts into one packaging and sorting application.

- Maintain a system run state from Start and Stop commands.
- Operate the conveyor only while the system is running and the batch is incomplete.
- Count packages detected by the package photoeye.
- Stop production automatically when 10 packages have been counted.
- Provide visual and audible batch-complete indication.
- Detect reject conditions and operate a timed reject diverter.
- Count completed reject cycles.
- Reset package count, reject count, batch-complete state, and system run state for the next batch.

3. Functional Requirements

- Start Push Button I:1/0 shall latch System_Run B3:0/0 only when Batch_Complete B3:0/2 is false.
- Stop Push Button I:1/1 shall unlatch System_Run.
- Conveyor Motor O:2/0 shall energize only when System_Run is true and Batch_Complete is false.
- System Running Lamp O:2/3 shall follow the same operating condition as the conveyor.
- Package Photoeye I:1/2 shall drive Package Counter C5:0 while System_Run is true.
- C5:0 shall have a preset of 10.
- When C5:0/DN becomes true, Batch_Complete B3:0/2 shall latch.
- Batch Complete Lamp O:2/4 and Buzzer O:2/2 shall energize while Batch_Complete is true.
- Reset Push Button I:1/4 shall reset the package counter and clear Batch_Complete and System_Run.
- Reject Sensor I:1/3 shall latch Reject_Cycle_Active B3:0/1 while System_Run is true.
- T4:0 shall time the reject cycle for two seconds using a 0.01-second time base and preset of 200.
- Reject Diverter O:2/1 shall energize while Reject_Cycle_Active is true.
- When T4:0/DN becomes true, Reject_Cycle_Active shall clear.
- Reject Counter C5:1 shall increment from the reject timer Done condition and shall have a preset of 999.
- Rung 0014 shall additionally reset C5:1 while repeating the package-counter, Batch_Complete, and System_Run reset actions already present in Rung 0008.

4. System Components

Component	Identification	Function
PLC Platform	Allen-Bradley MicroLogix 1100	Executes packaging/sorting logic
Start Push Button	I:1/0	Starts/latches System Run
Stop Push Button	I:1/1	Stops/unlatches System Run
Package Photoeye	I:1/2	Detects packages for counting
Reject Sensor	I:1/3	Starts reject cycle
Reset Push Button	I:1/4	Resets batch and counters
System Run	B3:0/0	Maintained operating state
Reject Cycle Active	B3:0/1	Maintained reject-cycle state
Batch Complete	B3:0/2	Latched batch-complete state
Conveyor Motor	O:2/0	Moves packages through station
Reject Diverter	O:2/1	Diverts rejected package
Buzzer	O:2/2	Audible batch-complete indication
System Running Lamp	O:2/3	Running indication
Batch Complete Lamp	O:2/4	Visual batch-complete indication
Package Counter	C5:0	Counts packages to preset 10
Reject Counter	C5:1	Counts completed reject cycles
Reject Timer	T4:0	Times two-second reject cycle

5. I/O and Data Assignment

Physical I/O

Device	Address	Type	Description
Start Push Button	I:1/0	Digital Input	Starts system
Stop Push Button	I:1/1	Digital Input	Stops system
Package Photoeye	I:1/2	Digital Input	Package counting event
Reject Sensor	I:1/3	Digital Input	Reject cycle trigger
Reset Push Button	I:1/4	Digital Input	Resets batch/counters
Conveyor Motor	O:2/0	Digital Output	Conveyor command
Reject Diverter	O:2/1	Digital Output	Reject command
Buzzer	O:2/2	Digital Output	Batch-complete buzzer
System Running Lamp	O:2/3	Digital Output	Run indication
Batch Complete Lamp	O:2/4	Digital Output	Batch-complete indication

Internal Bits, Counter, and Timer Assignment

Function	Address	Configuration / Purpose
System Run	B3:0/0	Maintained system state
Reject Cycle Active	B3:0/1	Timed reject-cycle state
Batch Complete	B3:0/2	Latched at package target
Package Counter	C5:0	CTU, PRE 10
Reject Counter	C5:1	CTU, PRE 999
Reject Timer	T4:0	TON, PRE 200, time base 0.01 s = 2.00 s

6. Control Strategy

I separated the program into system operation, package/batch handling, and reject handling.

The Start rung establishes System_Run only while Batch_Complete is false. Conveyor and System Running Lamp are then controlled from System_Run and XIO Batch_Complete, so reaching the package target automatically removes normal production outputs.

Package counting is performed by C5:0 from the Package Photoeye while the system is running. The counter Done bit latches Batch_Complete, which remains active until Reset.

The reject path latches Reject_Cycle_Active from the Reject Sensor. A two-second timer defines the diverter cycle. When the timer Done bit becomes true, the reject-cycle bit is unlatched and C5:1 increments.

Reset logic returns the system to a known condition. The current screenshots contain two reset rungs with overlapping functionality; I preserve both exactly as implemented.

7. Engineering Design Decisions

Design Decision	Reason
Used internal System_Run bit	Separated maintained system state from conveyor output
Blocked Start when Batch_Complete is active	Prevents immediate restart before batch reset
Used Batch_Complete as a conveyor interlock	Automatically stops production at target count
Latched Batch_Complete	Maintains completion indication until reset
Used both lamp and buzzer	Provides visual and audible completion indication
Used Reject_Cycle_Active latch	Maintains diverter cycle independent of a momentary sensor condition
Used two-second TON for reject cycle	Provides defined diverter operating duration
Counted rejects from timer completion	Counts completed reject cycles rather than raw sensor activation
Used separate reject counter C5:1	Provides reject quantity independent of package count
Preserved duplicate reset functionality	Keeps Version 1 faithful to current screenshots

8. Ladder Logic Description

The current ladder contains fifteen functional rungs, Rung 0000 through Rung 0014, followed by END at Rung 0015.

8.1 Rung 0000 - Start / Latch System Run

```
|----[ XIC I:1/0 ]----[ XIO B3:0/2 ]----- ( OTL B3:0/0 )----|
      Start_PB      Batch_Complete      System_Run
```

Start is accepted only while Batch_Complete is false.

8.2 Rung 0001 - Stop / Unlatch System Run

```
|----[ XIC I:1/1 ]----- ( OTU B3:0/0 )----|
      Stop_PB      System_Run
```

Stop removes the maintained System_Run state.

8.3 Rung 0002 - Conveyor Motor Control

```
|----[ XIC B3:0/0 ]----[ XIO B3:0/2 ]----- ( OTE 0:2/0 )----|
      System_Run      Batch_Complete      Conveyor_Motor
```

The conveyor runs only while the system is active and the batch is incomplete.

8.4 Rung 0003 - System Running Lamp

```
|----[ XIC B3:0/0 ]----[ XIO B3:0/2 ]----- ( OTE 0:2/3 )----|
      System_Run      Batch_Complete      System_Running_Lamp
```

The running lamp follows the same conditions as the conveyor.

8.5 Rung 0004 - Package Counting

```
|----[ XIC B3:0/0 ]----[ XIC I:1/2 ]-----[ CTU C5:0 ]----|
      System_Run      Package_Photoeye      Package_Counter    PRE: 10
```

C5:0 counts package events while System_Run is active.

8.6 Rung 0005 - Latch Batch Complete

```
|----[ XIC C5:0/DN ]-----[ OTL B3:0/2 ]----|
      Package_Counter_Done      Batch_Complete
```

When the package counter reaches 10, Batch_Complete latches.

8.7 Rung 0006 - Batch Complete Lamp

```
|----[ XIC B3:0/2 ]-----[ OTE 0:2/4 ]----|
      Batch_Complete      Batch_Complete_Lamp
```

The visual indication remains on while Batch_Complete is latched.

8.8 Rung 0007 - Batch Complete Buzzer

```
|----[ XIC B3:0/2 ]-----[ OTE 0:2/2 ]----|
      Batch_Complete      Buzzer
```

The buzzer remains on while Batch_Complete is latched.

8.9 Rung 0008 - Batch Reset

```
|----[ XIC I:1/4 ]-----+----[ RES C5:0 ]----|
      Reset_PB           |      Package_Counter
                        +----( OTU B3:0/2 )----|
                        |      Batch_Complete
                        +----( OTU B3:0/0 )----|
                        |      System_Run
```

Reset clears the package counter, Batch_Complete, and System_Run.

8.10 Rung 0009 - Start Reject Cycle

```
|----[ XIC B3:0/0 ]----[ XIC I:1/3 ]-----[ OTL B3:0/1 ]----|
      System_Run      Reject_Sensor      Reject_Cycle_Active
```

A reject detected while the system is running latches Reject_Cycle_Active.

8.11 Rung 0010 - Reject Cycle Timer

```
|----[ XIC B3:0/1 ]-----[ TON T4:0 ]----|
      Reject_Cycle_Active      Reject_Timer    PRE: 200    Time Base: 0.01
```

T4:0 provides a two-second reject cycle.

8.12 Rung 0011 - Reject Diverter

```
|----[ XIC B3:0/1 ]-----[ OTE 0:2/1 ]----|
      Reject_Cycle_Active      Reject_Diverter
```

The diverter remains energized throughout the reject cycle.

8.13 Rung 0012 - End Reject Cycle

```
|----[ XIC T4:0/DN ]------( OTU B3:0/1 )----|
      Reject_Timer_Done           Reject_Cycle_Active
```

When T4:0 completes, the reject-cycle latch is cleared.

8.14 Rung 0013 - Reject Counter

```
|----[ XIC T4:0/DN ]-----[ CTU C5:1 ]----|
      Reject_Timer_Done           Reject_Counter   PRE: 999
```

C5:1 increments from reject timer completion. The visible screenshots do not use C5:1/DN for a separate action.

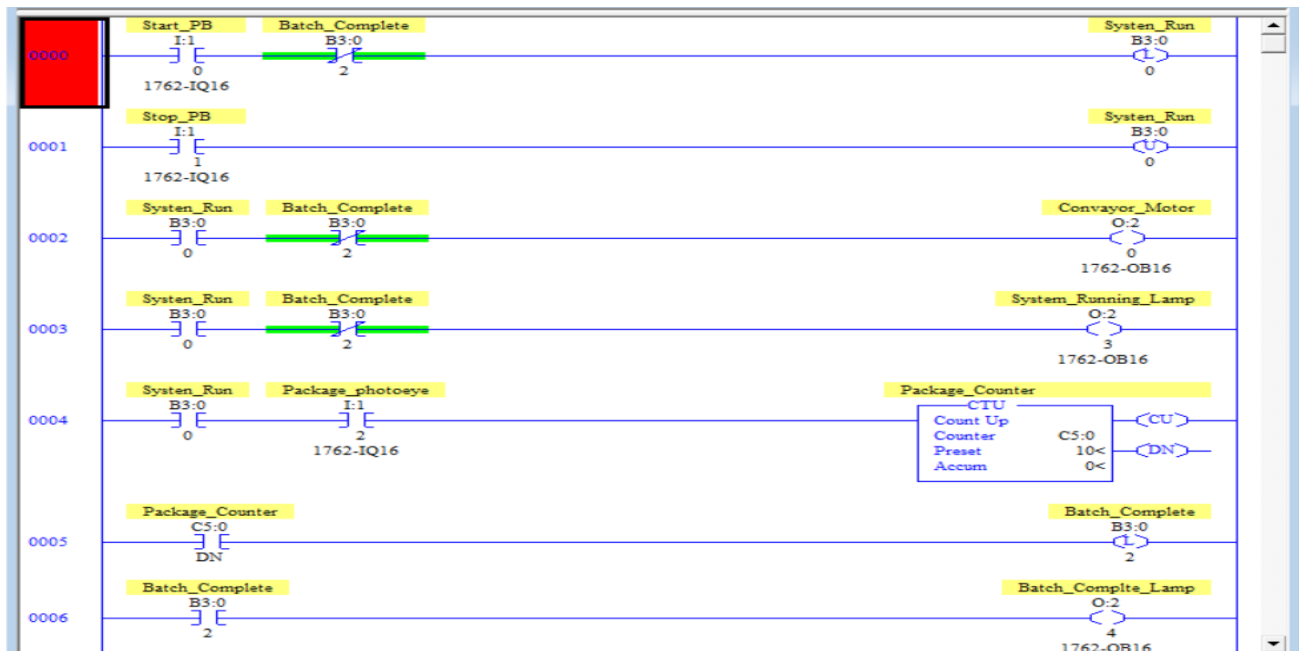
8.15 Rung 0014 - Full Counter / Batch Reset

```
|----[ XIC I:1/4 ]-----+----[ RES C5:0 ]----|
      Reset_PB              |   Package_Counter
                           +----[ RES C5:1 ]----|
                           |   Reject_Counter
                           +----( OTU B3:0/2 )----|
                           |   Batch_Complete
                           +----( OTU B3:0/0 )----|
                           |   System_Run
```

This second reset rung repeats the package/batch/run reset functions and additionally resets Reject Counter C5:1.

8.16 Actual RSLogix Implementation

The three figures below are the current RSLogix screenshots used as the technical source of truth for this report.



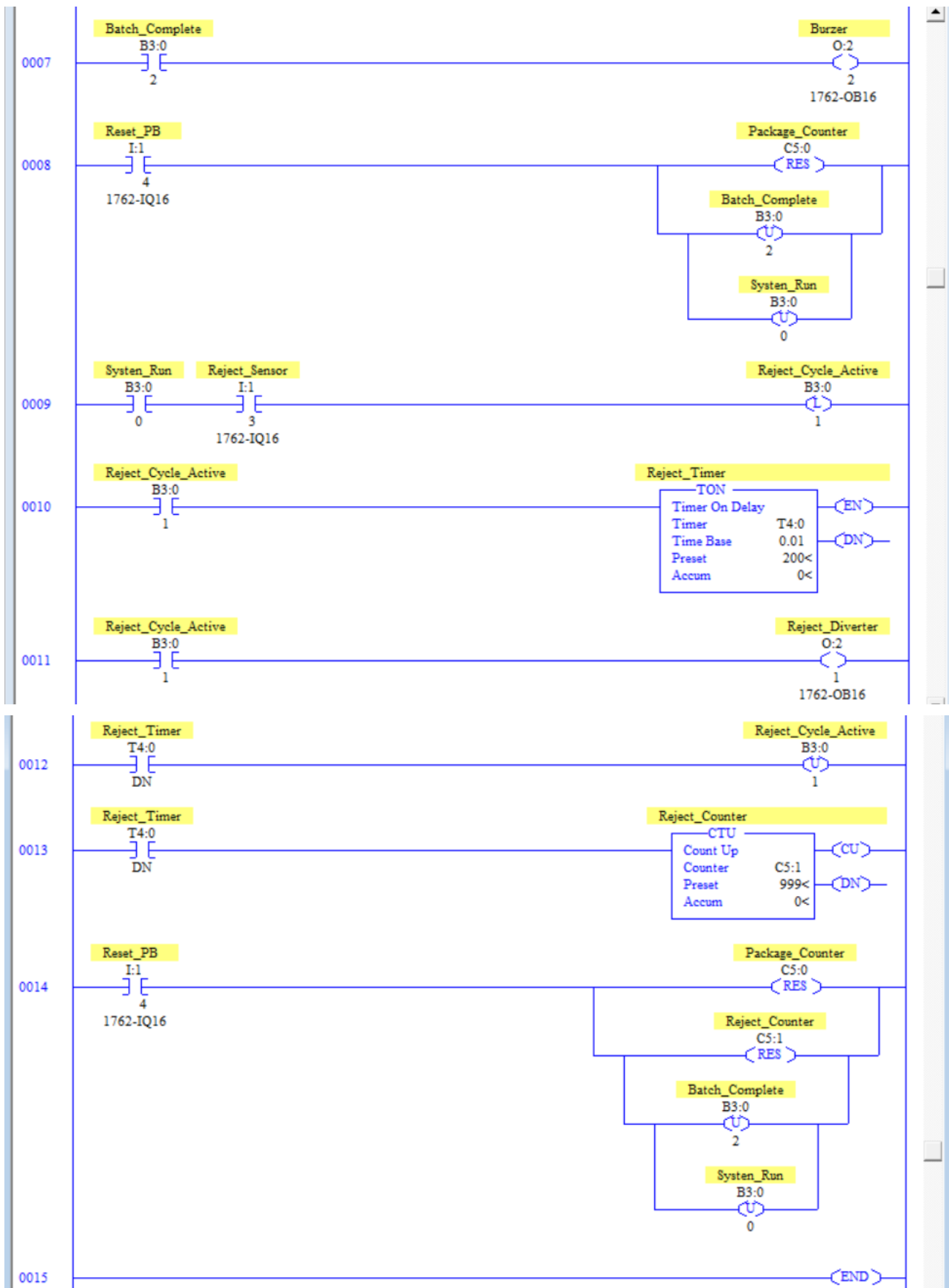


Figure P10.1 - RSLogix Packaging and Sorting Logic

9. Sequence of Operation

Step 1 - Start: Start latches System_Run while Batch_Complete is false.

Step 2 - Normal Production: Conveyor Motor and System Running Lamp energize.

Step 3 - Package Counting: Package Photoeye events increment C5:0.

Step 4 - Reject Detection: Reject Sensor latches Reject_Cycle_Active while System_Run is true.

Step 5 - Reject Diversion: Reject Diverter energizes and T4:0 times for two seconds.

Step 6 - Reject Cycle Complete: T4:0/DN clears Reject_Cycle_Active and increments C5:1.

Step 7 - Continue Production: Conveyor/package counting continues according to normal conditions.

Step 8 - Package Target Reached: At C5:0.ACC = 10, C5:0/DN latches Batch_Complete.

Step 9 - Batch Stop: Conveyor and System Running Lamp turn OFF.

Step 10 - Batch Indication: Batch Complete Lamp and Buzzer turn ON.

Step 11 - Reset: Reset clears package/reject counters through the visible reset rungs, clears Batch_Complete, and clears System_Run.

Step 12 - New Batch: A new Start command is required.

10. Testing & Validation

The current ladder screenshots represent the final tested Version 1 implementation. Tests below are derived only from behavior directly visible in the ladder and are recorded as PASS.

Test ID	Test Action / Condition	Expected Result	Observed Result	Status
T-01	Press Start with Batch_Complete false	System_Run latches	System_Run latched	PASS
T-02	System_Run true, batch incomplete	Conveyor and Running Lamp ON	Normal run outputs energized	PASS
T-03	Generate package event while running	C5:0 increments	Package count increased	PASS
T-04	Continue package events below 10	Batch_Complete remains false	Batch remained incomplete	PASS
T-05	Package count reaches 10	Batch_Complete latches	Batch_Complete latched	PASS
T-06	Batch_Complete true	Conveyor/Running Lamp OFF; Lamp/Buzzer ON	Batch stop/indication correct	PASS
T-07	Press Stop during normal running	System_Run clears	System stopped	PASS
T-08	Reject Sensor active while running	Reject_Cycle_Active latches	Reject cycle started	PASS
T-09	Reject cycle active	Diverter ON; T4:0 times	Diverter/timer operated	PASS
T-10	Reject timer reaches 2 s	Reject cycle clears	Reject cycle ended	PASS
T-11	Reject timer completes	C5:1 increments	Reject counter increased	PASS
T-12	Press Reset after batch completion	C5:0 reset; Batch_Complete/System_Run clear	Batch reset completed	PASS
T-13	Press Reset with reject count present	C5:1 resets through Rung 0014	Reject counter reset	PASS
T-14	Attempt Start while Batch_Complete still true	System_Run does not latch	Restart inhibited	PASS

Validation Summary

Testing confirmed normal start/stop operation, conveyor control, package counting, batch completion at 10 packages, automatic conveyor stop, visual/audible completion indication, timed reject-diverter operation, reject counting, reset behavior, and restart inhibition while Batch_Complete remains latched.

All fourteen defined Version 1 functional tests passed. The validation represents functional PLC project testing, not industrial commissioning or safety validation.

11. Results

The completed Industrial Packaging and Sorting Station successfully integrated conveyor operation, package counting, batch completion, reject sorting, operator indication, and reset behavior.

Package Counter C5:0 stopped the production sequence at a target of 10 packages by latching Batch_Complete. The conveyor and running lamp turned OFF while the Batch Complete Lamp and Buzzer turned ON.

The reject sequence operated for a timed two-second period and incremented Reject Counter C5:1 at completion.

All fourteen defined source-supported functional tests passed.

12. Challenges & Troubleshooting

A useful troubleshooting path is: System_Run -> Batch_Complete -> Conveyor/Running Lamp -> Package Photoeye -> Package Counter -> Reject Sensor -> Reject Cycle -> Reject Timer -> Diverter -> Reject Counter -> Reset.

Symptom	Possible Cause	Diagnostic Check	Corrective Action
System does not start	Batch_Complete true or Start condition false	Monitor I:1/0, B3:0/2, B3:0/0	Verify Rung 0000
Conveyor does not run	System_Run false or Batch_Complete true	Monitor B3:0/0 and B3:0/2	Verify Rung 0002
Package count does not increase	System_Run false or photoeye event absent	Monitor I:1/2 and C5:0.ACC	Verify package-count rung
Batch does not complete at 10	C5:0 preset/Done condition incorrect	Monitor C5:0.ACC and C5:0/DN	Verify counter and Rung 0005
Lamp/Buzzer not active at completion	Batch_Complete not latched or output rung issue	Monitor B3:0/2, O:2/4, O:2/2	Verify Rungs 0006/0007
Reject cycle does not start	System_Run false or Reject Sensor not active	Monitor B3:0/0, I:1/3, B3:0/1	Verify Rung 0009
Diverter stays ON too long	T4:0 not reaching Done or OTU not executing	Monitor T4:0 and B3:0/1	Verify Rungs 0010/0012
Reject count does not increase	Timer Done transition not seen by C5:1	Monitor T4:0/DN and C5:1.ACC	Verify Rung 0013
Reset leaves reject count	Only overlapping reset behavior partially executed	Monitor I:1/4, C5:1	Verify Rung 0014
System restarts unexpectedly	Batch_Complete may have been cleared	Monitor B3:0/2 and System_Run	Verify Start/Reset interaction

The project reinforced the value of tracing a sequence through internal states and counters. A problem at the final output often originates from an earlier permissive, latch, counter state, or timer condition.

13. Lessons Learned

This project helped me integrate multiple PLC control concepts into one larger application rather than treating each instruction as an isolated exercise.

I learned how a latched Batch_Complete state can be used as a machine-wide operating condition that stops the conveyor and changes operator indication.

The reject sequence demonstrated how a temporary sensor event can start a maintained timed action using an internal latch.

Counting rejects at timer completion showed that counters can represent completed process events rather than raw input activations.

The duplicate reset rungs also reinforced why larger programs benefit from later architectural cleanup and centralized reset handling.

14. Industrial Relevance

Packaging and sorting stations are common in manufacturing, food and beverage, pharmaceutical, logistics, warehouse, and consumer-goods production systems.

The Version 1 project demonstrates several general industrial concepts: maintained run control, conveyor interlocking, production counting, batch completion, reject diversion, timed actuators, and operator indication.

A real industrial system would normally require additional hardware feedback, safety circuits, sensor validation, reject verification, jam detection, fault handling, operating modes, controlled restart behavior, HMI diagnostics, and production reporting.

This Version 1 project should therefore be viewed as the foundation packaging/sorting implementation from which the industrial Version 2 design was later developed.

15. Transition to Industrial Version (Version 2)

Project 10 provides one of the clearest Version 1-to-Version 2 progressions in the portfolio. The foundation logic can be retained conceptually while the architecture is upgraded for stronger industrial operation.

Version 2 Improvement	Engineering Benefit
System Ready / Safety permissive architecture	Prevents operation unless required safe/healthy conditions are satisfied
Auto/Manual operating modes	Separates production and maintenance operation
Mode-conflict detection	Identifies invalid mode-command combinations
Run Request separation	Separates operator command from physical conveyor output
Conveyor permissive	Centralizes conveyor run conditions
Reject verification	Confirms that the rejected package actually reached the expected path
Reject-cycle timeout	Detects a reject sequence that fails to complete
Package sensor diagnostics	Detects stuck or abnormal package-photoeye behavior
Reject sensor diagnostics	Detects stuck or abnormal reject-sensor behavior
Quality warning	Provides process-quality indication based on reject activity
Master fault handling	Provides a common fault state and machine response
Controlled fault reset/recovery	Prevents uncontrolled restart after abnormal conditions
Controlled batch reset	Improves reset permissions and restart behavior
HMI integration	Provides machine status, alarms, counts, modes, manual controls, and diagnostics

A stronger Version 2 architecture can become: Start Request -> System Ready / Safety Checks -> Run Request -> Conveyor Permissive -> Package Processing -> Reject Detection -> Reject Cycle -> Reject Verification -> Counting / Quality Monitoring -> Batch Complete -> Controlled Reset, with fault handling and diagnostics surrounding the complete sequence.

The Version 2 design should preserve the recognizable packaging process while making machine state, abnormal conditions, and operator recovery explicit.

16. Project Summary

The Industrial Packaging and Sorting Station was the final Version 1 foundation project and combined maintained system operation, conveyor control, package counting, batch completion, reject diversion, reject counting, indication, and reset logic.

System_Run B3:0/0 controlled normal operation, C5:0 counted packages to a target of 10, and Batch_Complete B3:0/2 stopped the conveyor and activated the Batch Complete Lamp and Buzzer.

Reject Sensor I:1/3 started a latched Reject_Cycle_Active state. T4:0 timed the diverter for two seconds, after which the reject cycle cleared and C5:1 incremented.

The current screenshots also contain overlapping reset rungs, which are preserved exactly as part of the Version 1 source of truth.

All fourteen defined functional tests passed. This project provides a strong bridge from foundation PLC programming into the later industrial Version 2 packaging system with modes, permissives, diagnostics, reject verification, fault handling, safety-related shutdown behavior, and HMI integration.